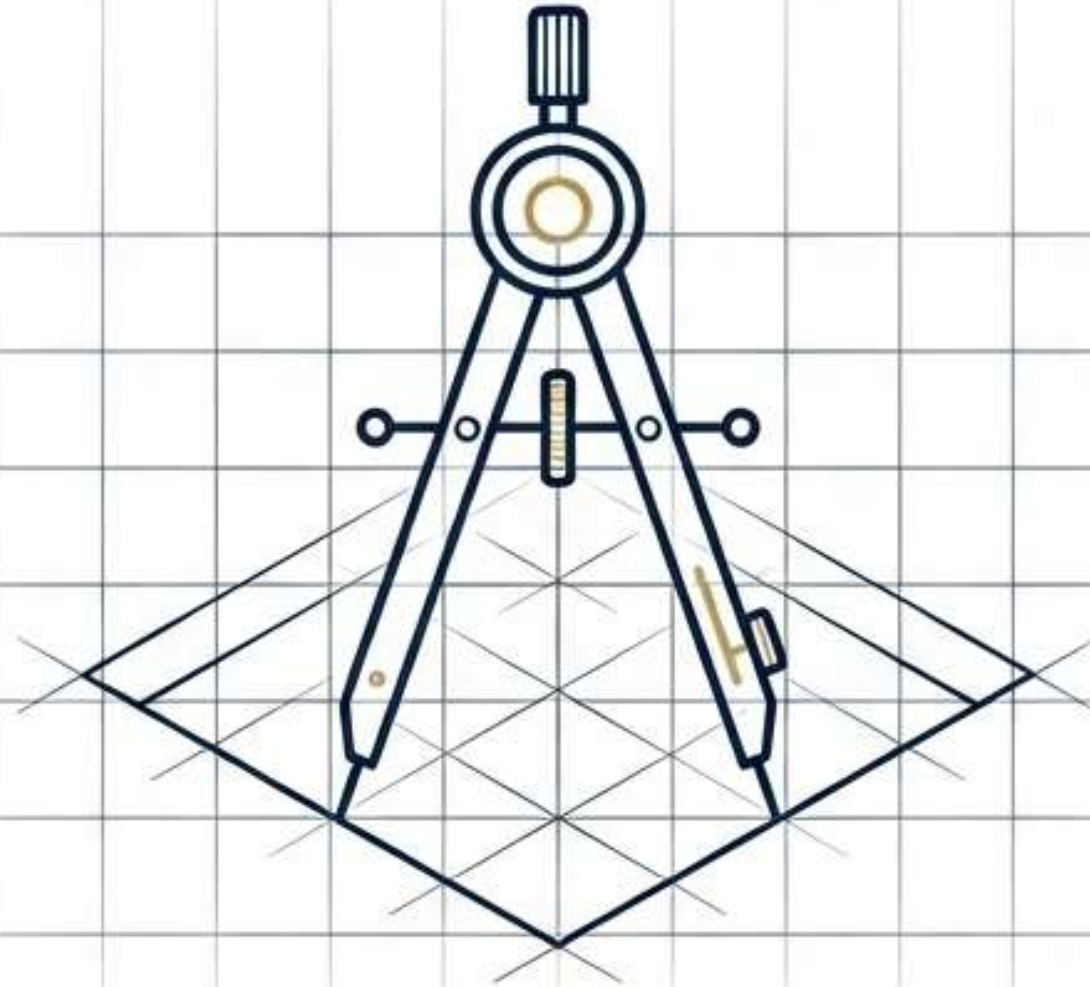




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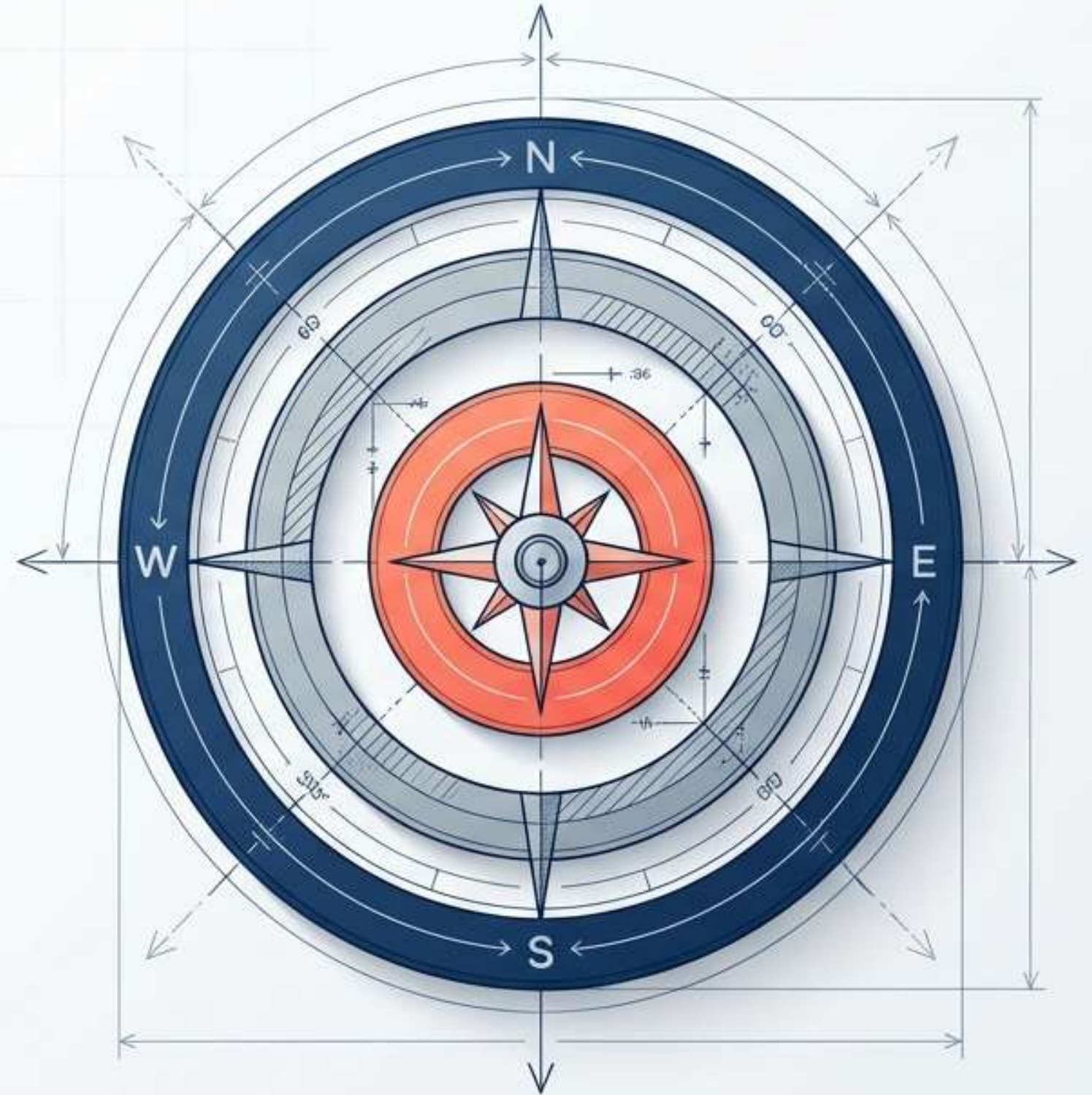


Topic 7: Project Risk Management

INS3044: IT Project Management

Learning Objectives

- ✦ Understand core risk concepts.
- ✦ Perform systematic risk analysis.
- ✦ Develop effective contingency plans.
- ✦ Navigate project uncertainty confidently.
- ✦ Apply frameworks to cases.
- ✦ Fulfill Course Learning Outcomes.



NAVIGATIONAL TARGET FRAMEWORK

The Concept of Risk

$$\text{Risk} = f(\text{Likelihood}, \text{Impact})$$

Projects are inherently uncertain.

Likelihood: Probability of occurrence.

Outcomes differ from plans.

Impact: Severity of consequence.

Manage unknown to survive.

Risk = Likelihood $\times$ Impact.

The Risk Management Process

Step 1: Identify Risks.

Step 2: Assess Risks.

Step 3: Plan Responses.

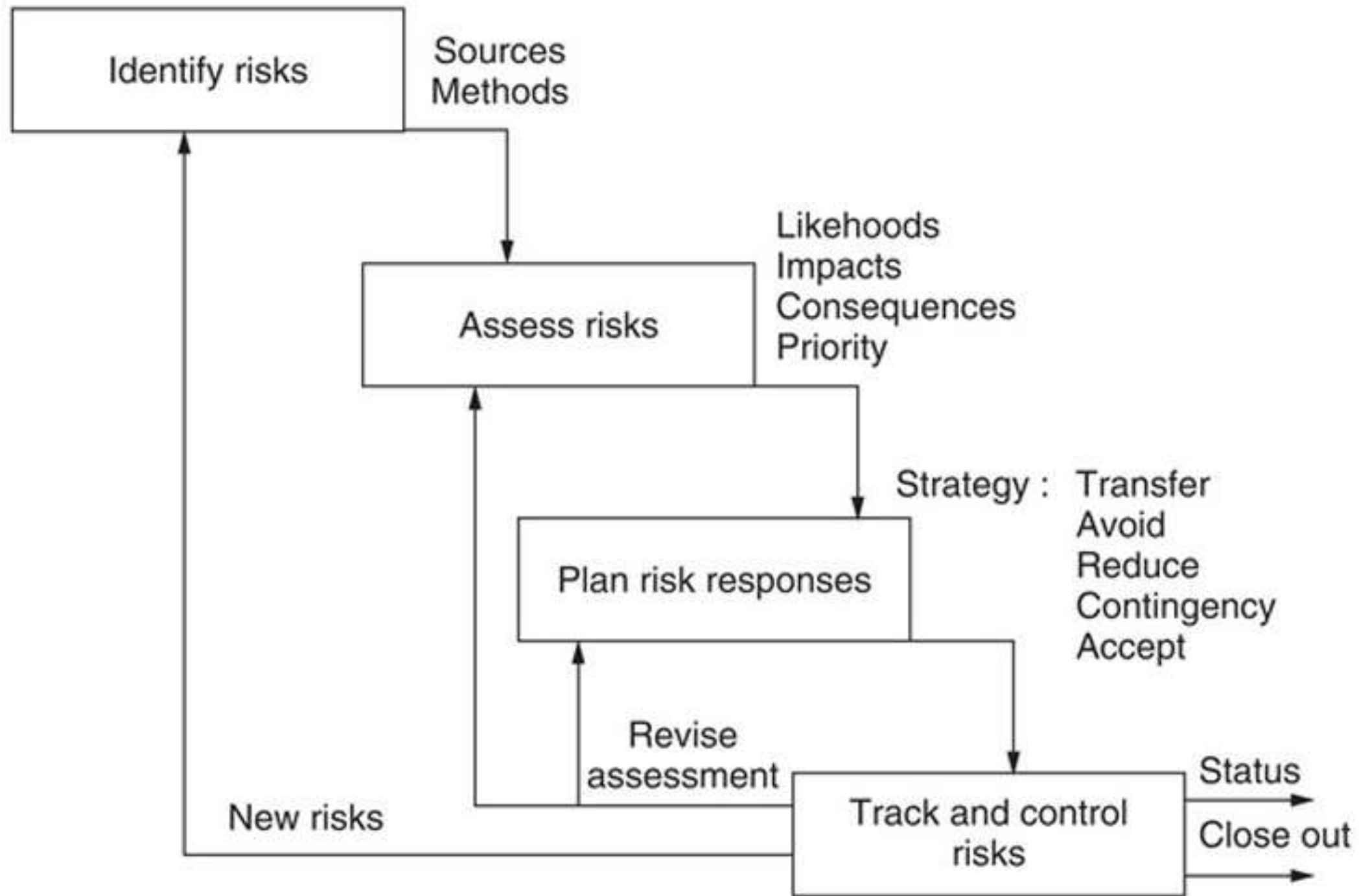
Step 4: Track Control.

Process is completely continuous.

Never stop managing risks.



The Risk Management Process (2)

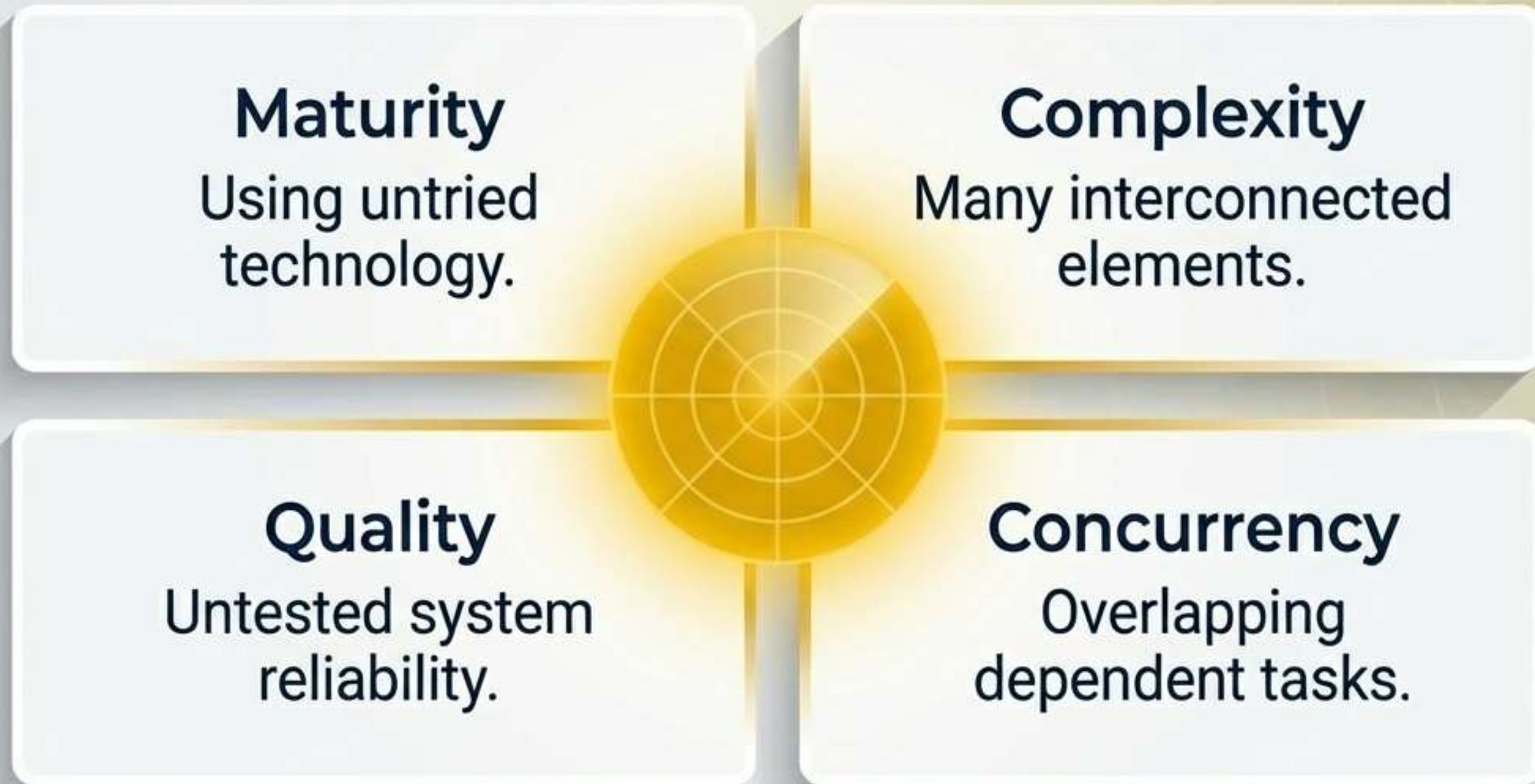


Step 1: Risk Identification

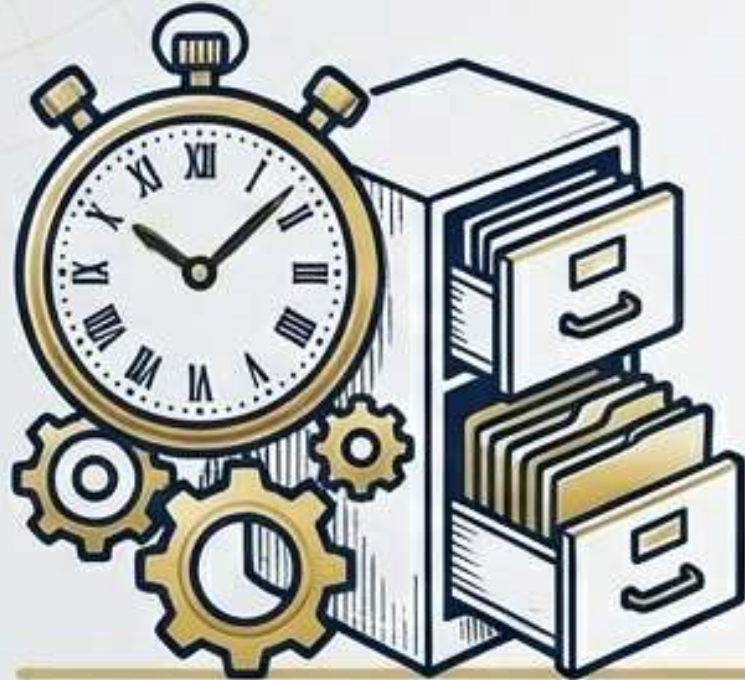
- Internal: Market needs fail.
- Internal: Technical problems arise.
- Internal: Faulty project assumptions.

- External: Competitors take action.
- External: Government regulations change.
- External: Natural environment hazards.

High-Risk IT Factors



Tools: Historical & Structural



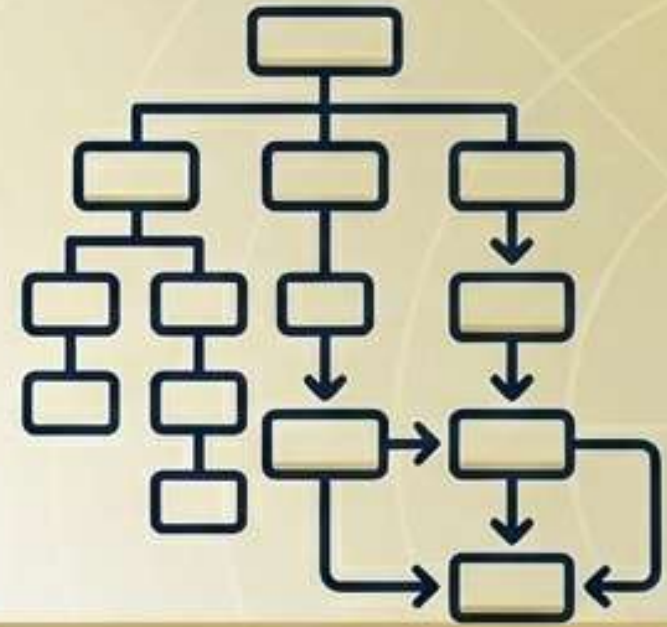
Analogy

- Learn from past projects.



Checklists

- Update from prior experience.



WBS & Flowcharts

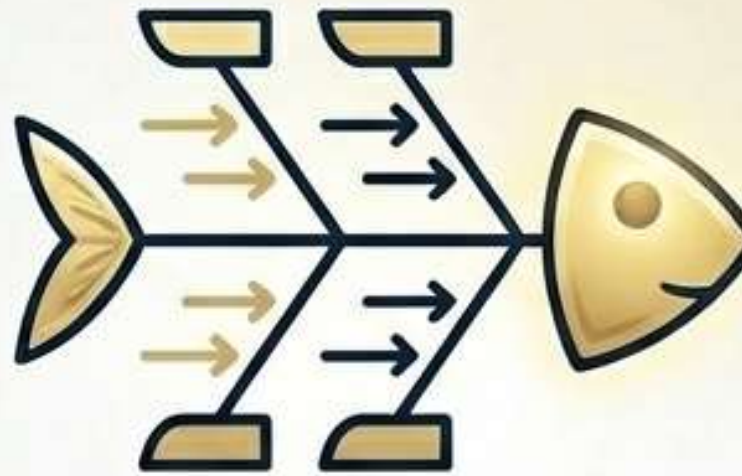
- Scrutinize every work package.
- Pinpoint process trouble spots.

Tools: Creative & Expert



Brainstorming

- Share team opinions.
- **Rule:** No assessing during brainstorming.



Cause-Effect

- Map sources to outcomes.

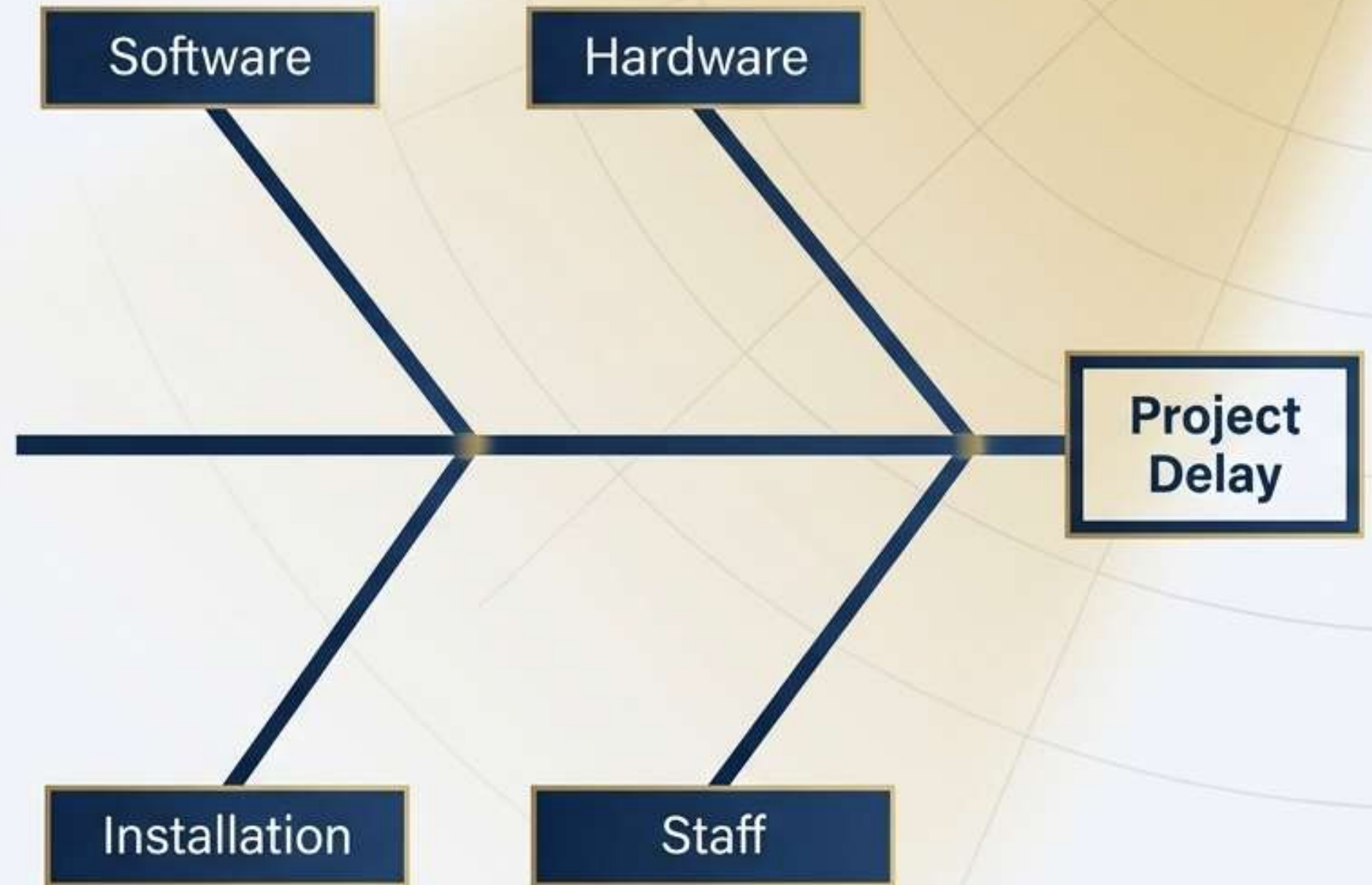


Delphi Technique

- Anonymous expert surveys.

Cause-and-Effect Diagram

- Also called Ishikawa diagram.
- Maps causes to specific effects.
- Categorizes fundamental risk sources.
- Highly visual team exercise.



App Lab 1: Brainstorming



Scenario: **QuickBite** deployment.

- Timer: Three minute sprint.

Identify two internal risks.

Identify two external risks.

Apply the Fishbone method.

Share with the class.

Step 2: Risk Assessment

- Likelihood: Probability of occurrence.
- Impact: Severity of damage.
- Measured in time delay.
- Measured in cost overrun.
- High risk demands attention.
- Calculate Expected Value consequences.

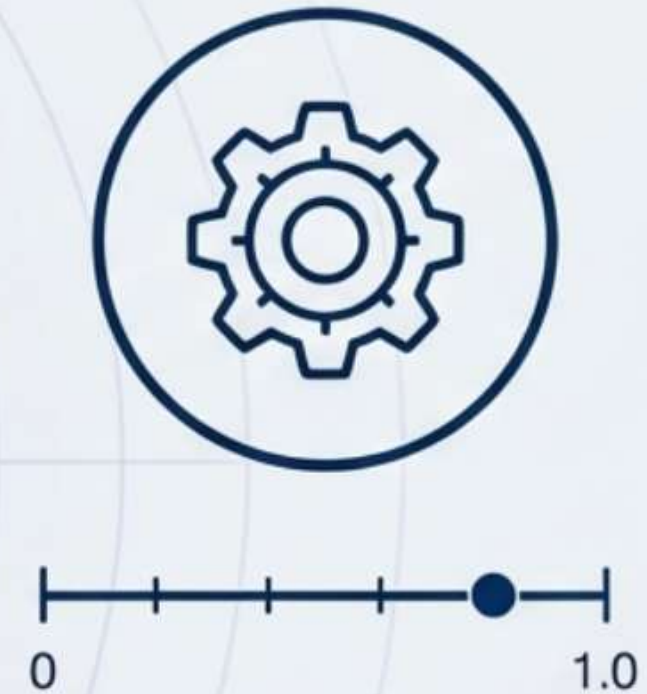
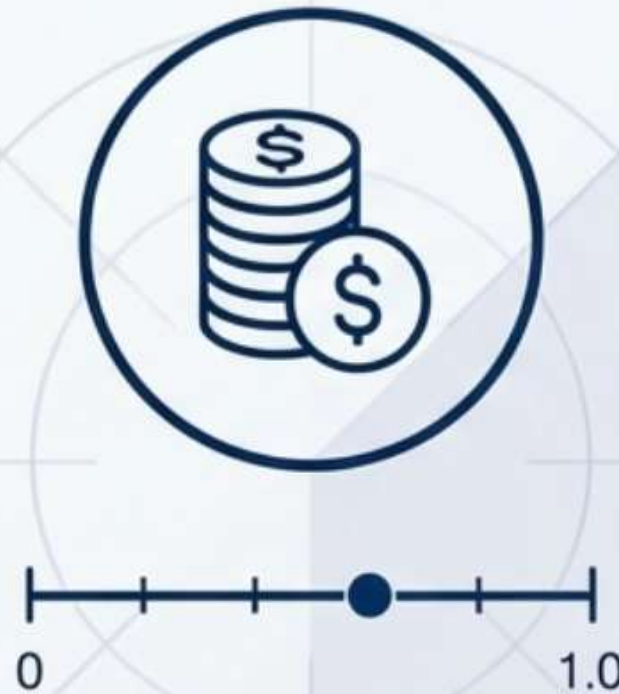


Dimension 1: Risk Likelihood

- Probability of a risk materializing.
- Measured on 0 to 1.0 scale.
- Can use qualitative rating scales.
- Common ratings: High, Medium, Low.
- Estimates rely on expert judgment.
- Combine opinions for better accuracy.



Dimension 2: Risk Impact



- Consequences if risk actually happens.
- Severely affects time, cost, performance.
- Measured on 0 to 1.0 scale.
- Rated as High, Medium, Low.
- Assess without planned preventive measures.
- Evaluate worst-case scenario impacts.

Practice & Discussion 1



Scenario: New software system integration.

1. **Identify** one potential technical risk.
2. **Estimate** qualitative likelihood (H/M/L).
3. **Estimate** qualitative impact (H/M/L).



Discuss reasoning with your peers.

Time limit: 3 minutes.

Calculating Risk Consequence

Probability (0 to 1.0)

Severity (0 to 1.0)

$$\text{RCR} = \text{Likelihood} \times \text{Impact}$$

Risk Consequence Rating

- Combines likelihood and impact together.
- RCR ranges from 0 to 1.0.
- High RCR demands immediate attention.
- Used to compare and prioritize.

Expected Value Consequences

Translate risks into real numbers. Helps calculate necessary project buffers.



Expected Risk Time (RT)

$RT = \text{Estimated Delay} \times \text{Likelihood.}$



Expected Risk Cost (RC)

$RC = \text{Estimated Cost} \times \text{Likelihood.}$

Practice & Discussion 2



Scenario: Hardware shipment delay risk.

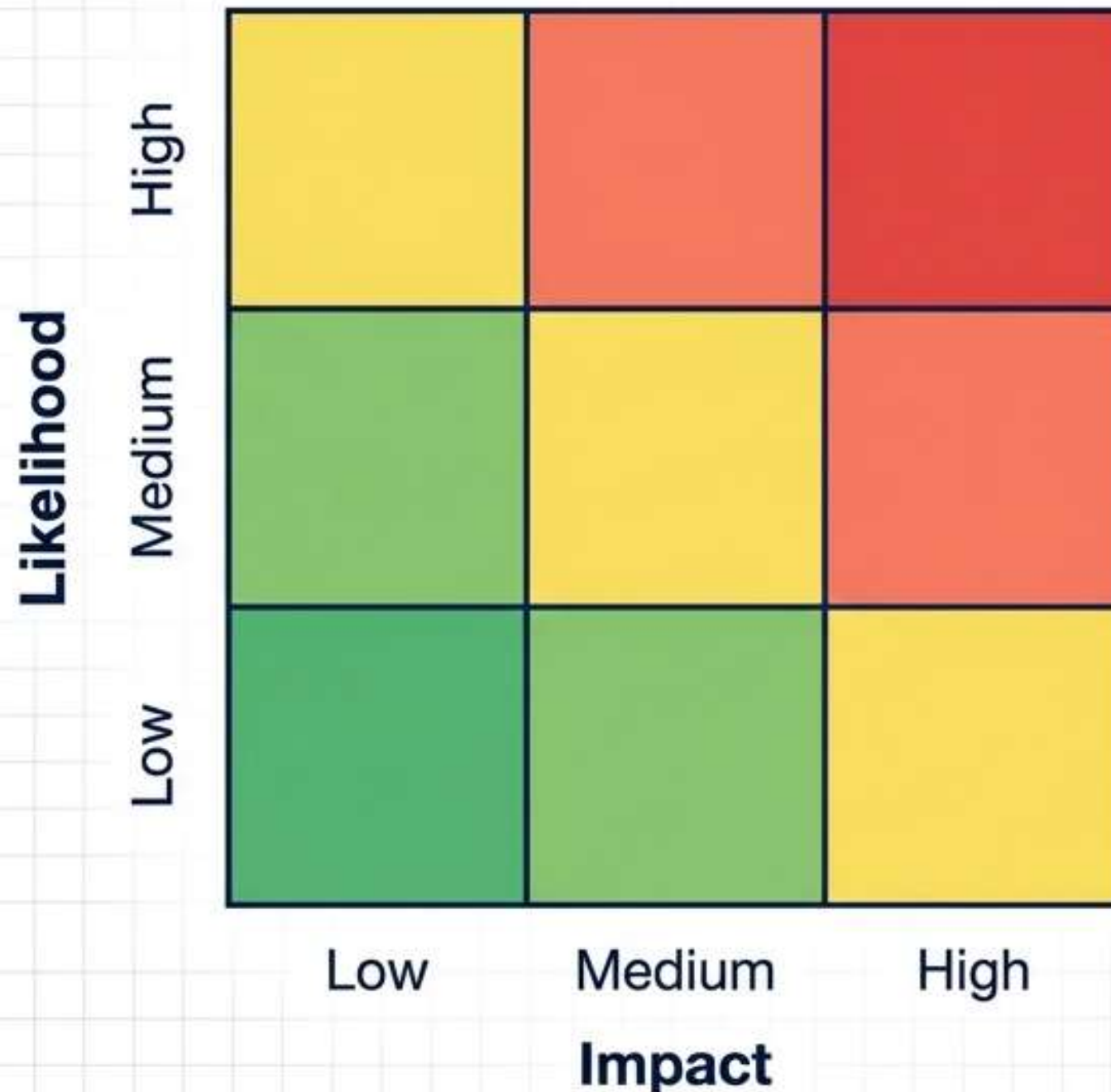
Likelihood of delay: 40% (0.4).

Estimated delay impact: 5 weeks.

Estimated cost impact: \$10,000.

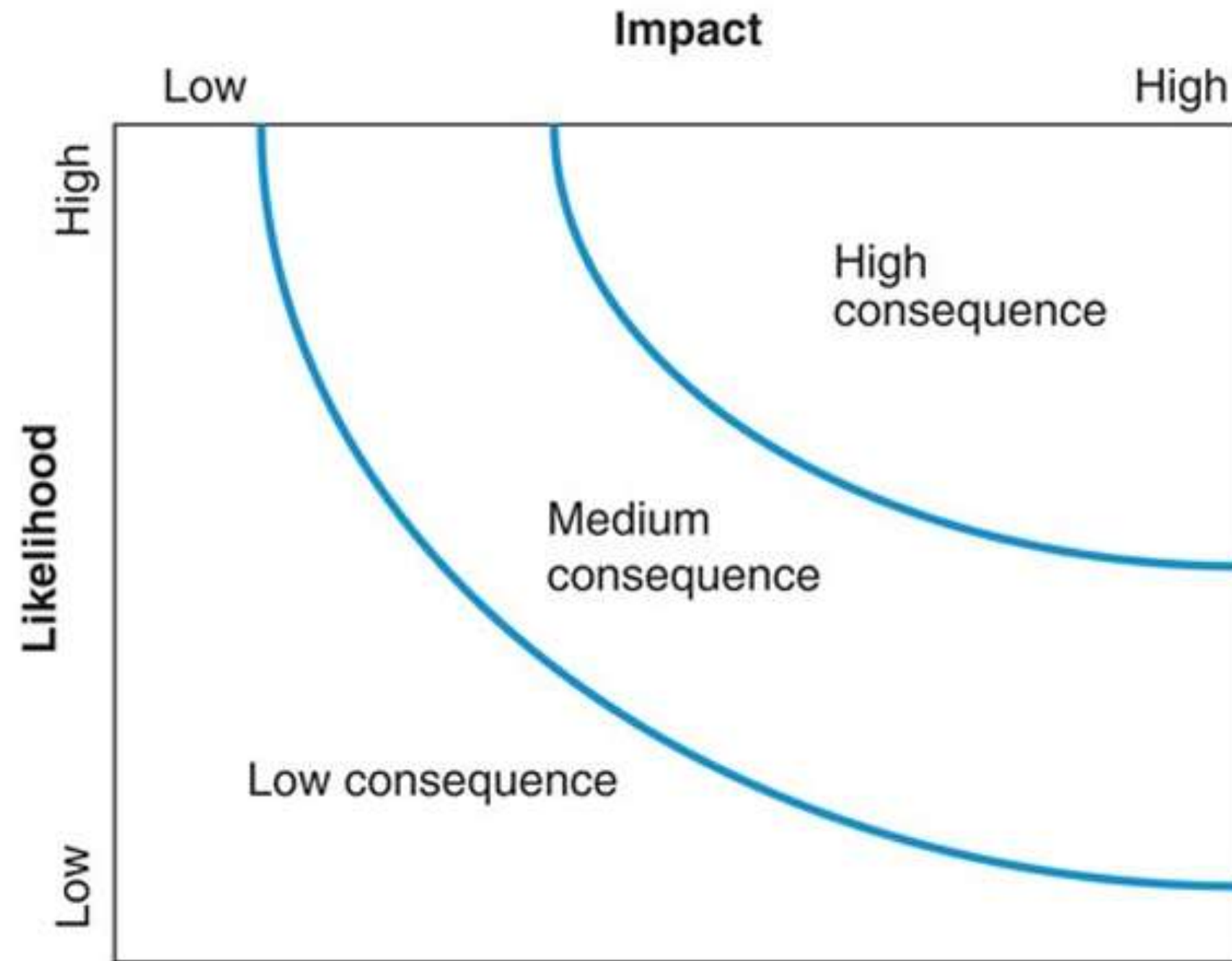
-
- Calculate Expected Risk Time (RT).
 - Calculate Expected Risk Cost (RC).

The Risk Matrix Heatmap



- Plot likelihood versus impact.
- Low consequence (Green zone).
- Medium consequence (Yellow zone).
- High consequence (Red zone).
- Prioritize top right risks.
- Record in risk register.

Risk Consequence



The Risk Register Framework

- Log all identified project risks.
- Sort by highest consequence rating.
- Focus on moderate-to-high risks.
- Assign specific risk owners.
- Develop targeted mitigation strategies.
- Never ignore severe impact risks.



	Risk ID	Impact	Likelihood	Consequence Rating	Mitigation Action
●	—	—	—	—	—
●	—	—	—	—	—
●	—	—	—	—	—
●	—	—	—	—	—

App Lab 2: Quantification

$$\text{Expected Risk Cost} = \text{Likelihood} \times \text{Cost Impact}$$



Scenario: Hardware shipment delay.

Likelihood of delay: 30%.

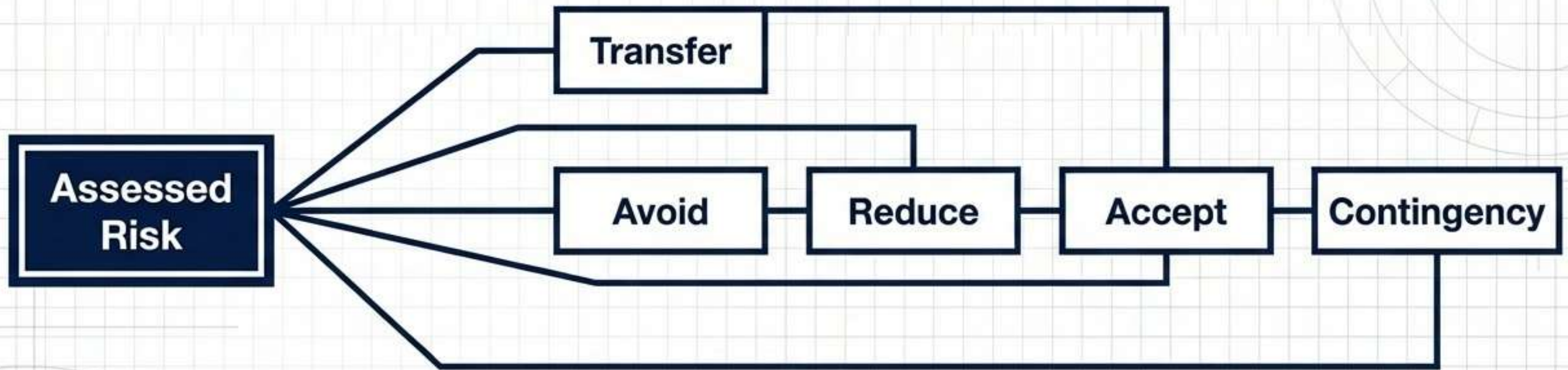
Delay cost impact: \$10,000.

Formula: $RC = L \times I$.

Calculate Expected Risk Cost.

What is your answer?

Step 3: Response Planning



- Transfer: Shift to others.
- Avoid: Change project plan.
- Reduce: Lower likelihood impact.
- Accept: Acknowledge and proceed.
- Contingency: Plan backup actions.
- Match strategy to risk.

Strategy Group 1: Proactive Action



AVOID:

- Eliminate the risk source entirely.
- Change project scope or schedule.



REDUCE:

- Lower likelihood or impact level.
- Test early, use better resources.



Practice 1: Avoid or Reduce?

Scenario A:

Drop unstable software feature completely.



Scenario B:

Run extensive beta testing phases.





Practice 1: Avoid or Reduce?

Scenario A:

Drop unstable software feature completely.



(Answer: AVOID)

Scenario B:

Run extensive beta testing phases.



(Answer: REDUCE)

Strategy Group 2: Shifting Burden



Strategy Group 3: Reality Check



ACCEPT:

Do nothing proactively.
Cost of fixing exceeds benefit.



CONTINGENCY:

Create a backup Plan B.
Execute only if trigger occurs.



Practice 2: Which Strategy?

Context: Server delivery might delay.

1. Buy late-penalty insurance?

2. Have cloud backup ready?

3. Tolerate a 1-day delay?



Practice 2: Which Strategy?

Context: Server delivery might delay.

1. Buy late-penalty insurance?

(Transfer)

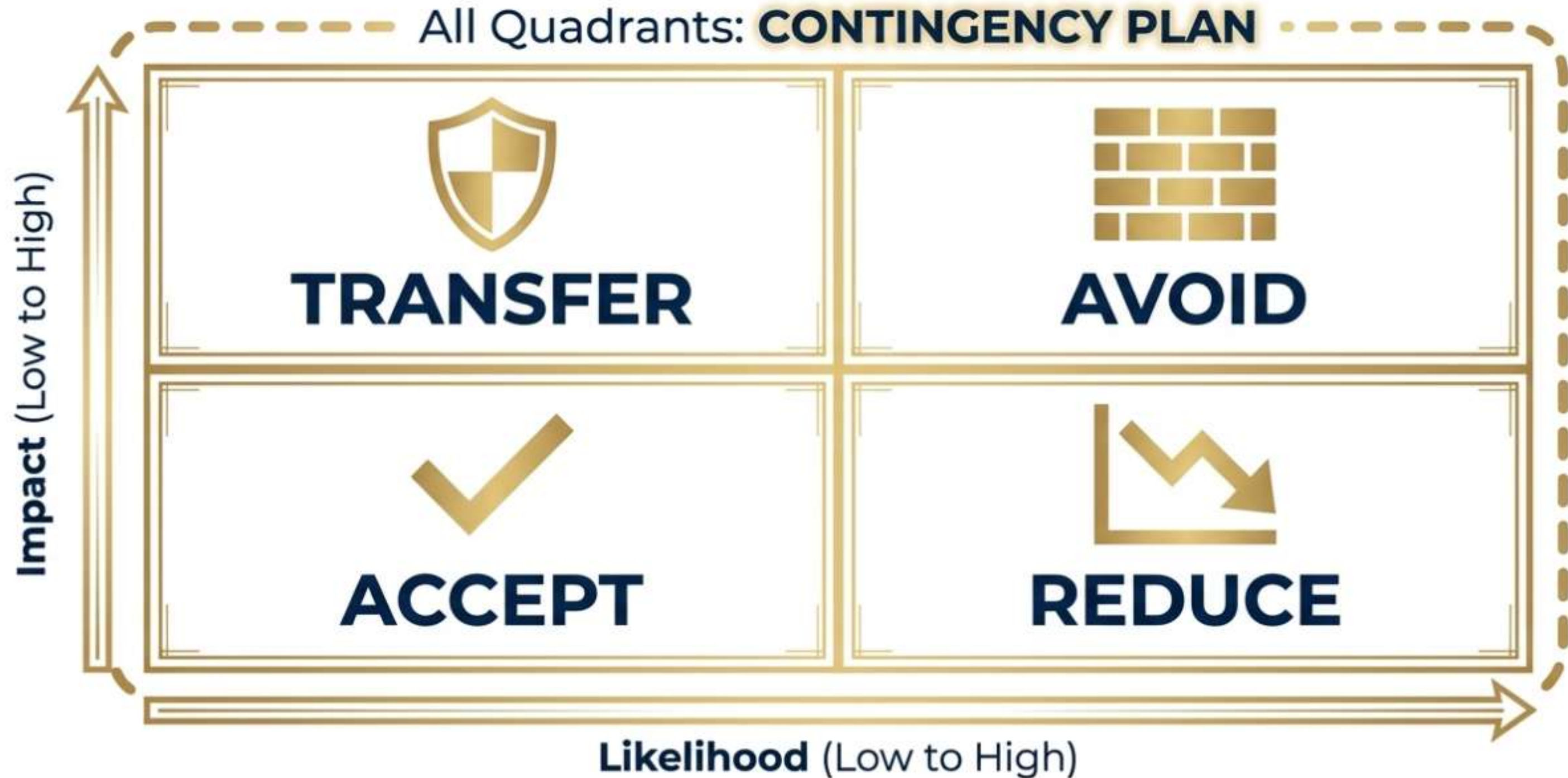
2. Have cloud backup ready?

(Contingency)

3. Tolerate a 1-day delay?

(Accept)

The Response Decision Matrix



App Lab 3: Strategy Match

Scenario: Critical server crash.

Project Actions

Action A: Buy insurance.

Action B: Install generator.

Action C: Cancel upgrade.

Action D: Do nothing.

Strategies

Strategy: Avoid

Strategy: Accept

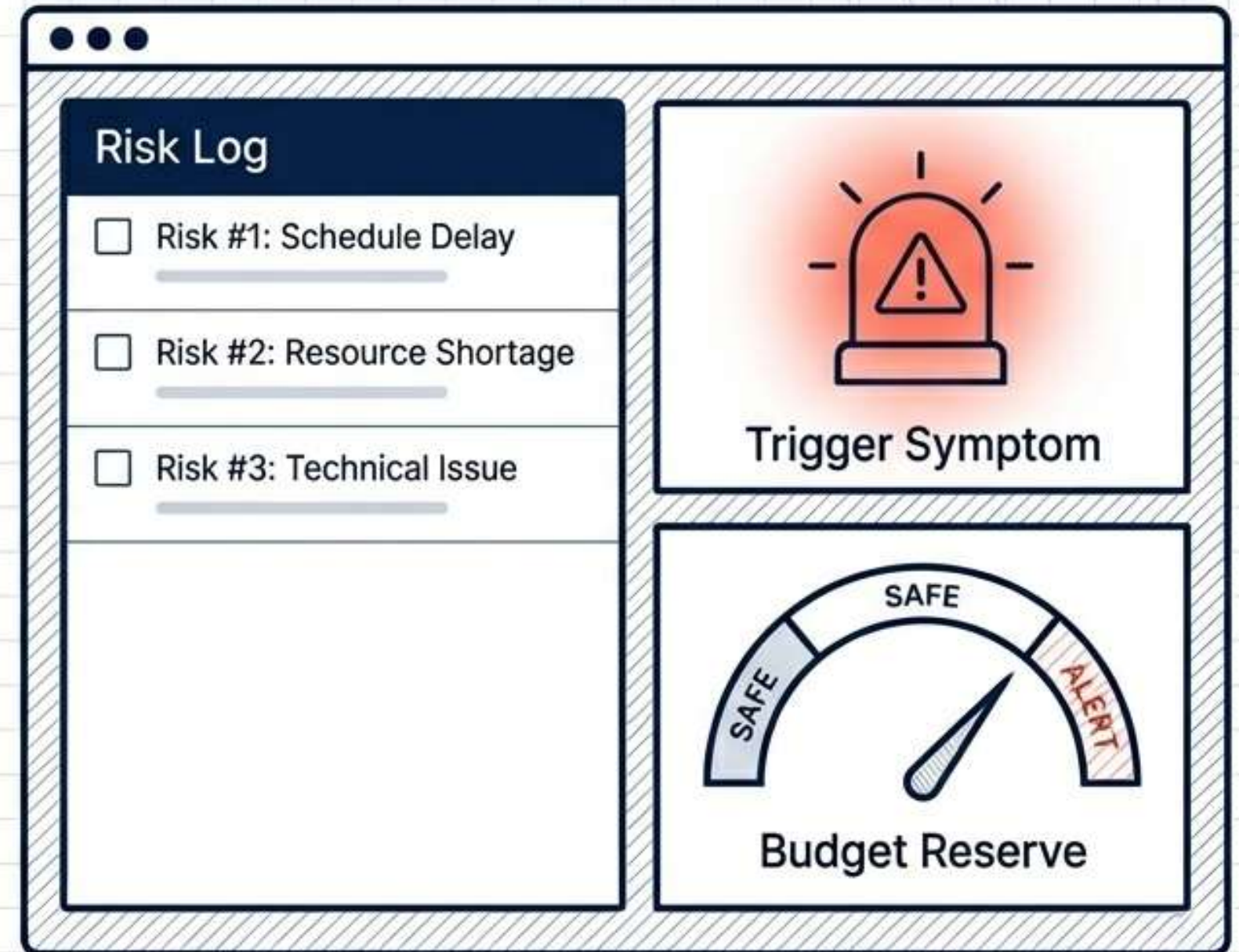
Strategy: Transfer

Strategy: Reduce

Match actions to strategies.

Step 4: Track & Control

- Document in risk log.
- Watch for trigger symptoms.
- Execute planned contingency actions.
- Hold schedule time reserves.
- Hold budget money reserves.
- Expect the unexpected always.



Takeaways & Final Discussion

Identify, Assess, Plan, Track.
Risk = Likelihood × Impact.
Prepare contingency action plans.

1

Q1: Why avoid risk avoidance?

2

Q2: How to spot triggers?

3

Q3: Who owns the risk?